

Reflection Log Semester Average

How did you approach understanding the problem?

-I began by examining the program's requirements, which included allowing the user to input student data, compute the semester average, and store or see the outcomes. To determine exactly what needed to be developed, I examined each feature and Since stats part2 was comparable to this mastery, I used the Stats Part 2 skillbuilder and I was able to add features and do touchups on my mastery from that skillbuilder after seeing that it was very similar to this mastery.

Were there any parts of the problem you found confusing at first? If so, how did you resolve that confusion?

-At first I found it confusing how to properly handle user input, especially when the user leaves fields empty or types letters instead of numbers. So I revised my Stats Part 2 code and made touches to my Semester Average Mastery. and That helped me try running the program and saw what errors appeared. From there, I fixed them one by one.

Planning the solution

Did you create a plan or break the problem into smaller steps before coding?

-Yes, I separated the program into small steps: setting up the GUI, planning my design, getting input, calculating the grade average, saving to the file, and viewing the file.

How did you decide on the tools, data structures, or algorithms to use?

-I used simple Swing components (JTextField, JButton, JTextArea) because they were required and easy to work with. I used a normal text file for saving since it kept things simple.

Implementation

Did you write the code in small pieces or attempt the entire solution at once?

-I wrote the code in small pieces. I tested each button right after writing it so I could fix problems quickly.

How did you test your solution along the way to make sure it was working?

-After adding each feature, I ran the program and tried entering data, leaving fields blank, and clicking the buttons to check that everything worked.

Overcoming Challenges

What part of the problem was the most difficult for you?

-The hardest part was handling input errors correctly. When the user entered letters instead of numbers, the program crashed at first.

How did you handle moments when you felt stuck or unsure of what to do next?

-I looked at the error message, searched up what the error meant, and then fixed the problem by adding error checks or simple try/catch statements. Sometimes I would ask my buddies around me for feedback or help. or simple try/catch statements. I also checked for similarities in my skill builders and that helped me and guided me through when I felt stuck or I felt unsure.

Learning

Was there anything you learned that you think will help you with future challenges?

-I learned that testing as you go makes everything easier. I also learned how important it is to handle errors so the program doesn't crash. These skills will definitely help with bigger Java projects.